

§6: Estimation and Testing Linear Combinations of Group Means

• Spock Example:

$$y_{ij} = \mu_i + \varepsilon_{ij}$$

↑

90 women in versus mean 90 women
of judge i for judge i

individual specific noise

μ_i = Spock's judge

- Want to compare μ_i to $\mu_2, \dots, \mu_7$ without having to assume $\mu_2 = \dots = \mu_7$

$$H_0: \mu_1 = \frac{1}{6}(\mu_2 + \dots + \mu_7)$$

$$H_A: \mu_1 \neq \frac{1}{6}(\mu_2 + \dots + \mu_7)$$

$$\Rightarrow \mu_1 = \frac{1}{6}\mu_2 + \frac{1}{6}\mu_3 + \dots + \frac{1}{6}\mu_7$$

$$\Rightarrow \mu_1 - \frac{1}{6}\mu_2 - \frac{1}{6}\mu_3 - \dots - \frac{1}{6}\mu_7 = 0$$

this is a linear combination of the means

- Def: A linear combination of $\mu_1, \mu_2, \dots, \mu_I$ with coefficients $c_1, c_2, \dots, c_I$ is

$$c_1\mu_1 + c_2\mu_2 + \dots + c_I\mu_I$$

If $c_1 + c_2 + \dots + c_I = 0$, then this linear combination is called a contrast.

- Let $\delta = \mu_1 + \left(-\frac{1}{6}\right)\mu_2 + \left(-\frac{1}{6}\right)\mu_3 + \dots + \left(-\frac{1}{6}\right)\mu_7$

$$\uparrow C_1 = 1, \quad C_2 = C_3 = \dots = C_7 = -\frac{1}{6}$$

- In general $\delta = C_1\mu_1 + C_2\mu_2 + \dots + C_I\mu_I$

$$H_0: \delta = 0$$

$$H_A: \delta \neq 0$$

- Estimate δ by replacing each μ_i with $\bar{Y}_{i\cdot}$

$$\hat{\delta} = C_1\bar{Y}_{1\cdot} + C_2\bar{Y}_{2\cdot} + \dots + C_I\bar{Y}_{I\cdot}$$

- For large n , $\hat{\delta} \sim N(\delta, \text{var}(\hat{\delta}))$,

$$\text{where } \text{var}(\hat{\delta}) = \sigma^2 \left(\frac{C_1^2}{n_1} + \frac{C_2^2}{n_2} + \dots + \frac{C_I^2}{n_I} \right)$$

- under H_0 : $\hat{\delta} \sim N(0, \text{var}(\hat{\delta}))$

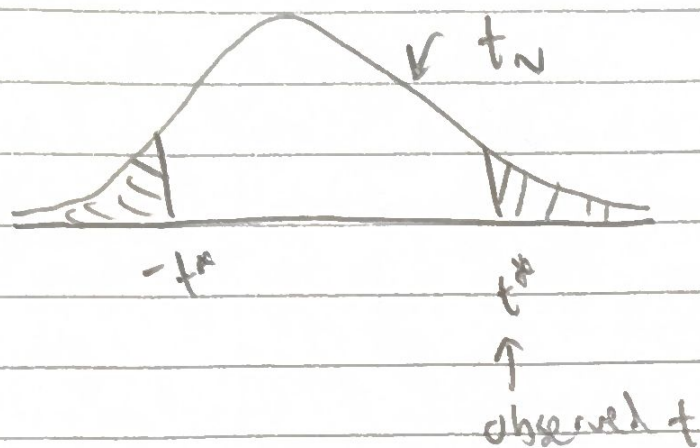
$$\Rightarrow \frac{\hat{\delta}}{\text{se}(\hat{\delta})} \sim N(0, 1)$$

The Standard Error of $\hat{\delta}$ is $sp \sqrt{\frac{C_1^2}{n_1} + \frac{C_2^2}{n_2} + \dots + \frac{C_I^2}{n_I}} = \text{SE}(\hat{\delta})$

$$\frac{\hat{\delta}}{\text{SE}(\hat{\delta})} \sim t_N$$

$$\text{where } N = n_1 + n_2 + \dots + n_I - I$$

- Get a p-value for testing $H_0: \gamma = 0$ vs $H_A: \gamma \neq 0$



$$pt(-abs(t^*), df = n)$$

- Get a 95% CI for γ

$$\hat{\gamma} \pm t_n(0.975) SE(\hat{\gamma})$$

- In R, it assumes parameterization of the form

$$Y_{1j} = \mu + \epsilon_{1j}$$

$$Y_{2j} = \mu + \delta_2 + \epsilon_{2j}$$

$$Y_{3j} = \mu + \delta_3 + \epsilon_{3j}$$

⋮

$$Y_{Ij} = \mu + \delta_I + \epsilon_{Ij}$$

It returns coefficient estimates

$$coef(aout) = \hat{\mu}, \hat{\delta}_2, \hat{\delta}_3, \dots, \hat{\delta}_I$$

$$\begin{array}{lcl} \mu_1 = \mu & & \mu = \mu_1 \\ \mu_2 = \mu + \delta_2 & \Rightarrow & \delta_2 = \mu_2 - \mu_1 \\ \vdots & & \vdots \\ \mu_i = \mu + \delta_i & & \delta_i = \mu_i - \mu_1 \end{array}$$

• To force R to use $\mu_1, \mu_2, \dots, \mu_i$ parameterization, use

`aov(Response ~ Exploratory - 1, data = dt)`

↑ removes an intercept term

I But only do this just before running `linearHypothesis()`

07 - linear combos in R, p14

08 - simultaneous inference, p14